

Popy Halder

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Education

B.S. Cybersecurity, University at Albany

May 2026

Relevant coursework: Trustworthy AI, Cybersecurity Basics, Networking and Cryptography, Information Security Privacy, Cybersecurity Long-Term Planning & Risk Management, Programming for Problem-Solving (Python), Introduction to Web Technologies (HTML/CSS/JavaScript), Introduction to Networks and Systems, Database Fundamentals for Security

Experience

Consultant Intern — Cyber Risk & Third-Party Risk Management

January - April 2025

Ernst & Young Trajectory Program, Albany, NY

Tools:

- Conducted a third-party cybersecurity risk assessment using NIST CSF, NIST SP 800-53, NIST RMF, and FFIEC guidance to evaluate vendor security and compliance risks.
- Developed inherent and control questionnaires and a likelihood-severity risk-scoring methodology to quantify inherent and residual risk across data breach, vendor security, regulatory, operational, and insider-threat scenarios.
- Designed TPRM implementation plans for 1,500+ clients, using cost-benefit analysis to prioritize security controls, remediation efforts, and phased implementation.
- Evaluated security controls and proposed PAM/identity-security and SLA-monitoring solutions to strengthen vendor oversight and reduce third-party exposure.
- Developed risk mitigation recommendations that reduced modeled total residual risk by approximately 42%, including a 75% reduction in weak vendor-oversight risk and 56% reduction in insider-threat risk.

Cybersecurity Intern — Security Operations & Controls Assurance

July – August 2023

NYC DOE – Division of Instructional & Information Technology, Brooklyn, NY

- Executed DMZ security and asset validation, correlating network assets to support vulnerability management and threat mitigation.
- Audited VPN remote access using Cisco ISE, validating authentication, user access, and security controls.
- Completed hands-on Zscaler ZIA, ZPA, and ZDX training, applying Zero Trust, secure internet access, and private application access concepts.
- Managed endpoint asset tagging in CrowdStrike and performed Rapid7 vulnerability scans to identify and prioritize security risks.

Projects

Researcher

December 2025

A Configurable Guardrail System for Large Language Models: A Firewall-Based Approach to Mitigating Malicious Prompts

University at Albany

- Co-authored research on a firewall-based LLM guardrail architecture for mitigating prompt injection,

jailbreaking, adversarial prompts, and data exfiltration.

- Designed a multi-layered input/output guardrail architecture incorporating IOC-based rules, signature-based detection, semantic analysis, PII detection, and configurable security policies.
- Developed an evaluation methodology using adversarial testing, TPR, FPR, accuracy, and latency benchmarking to evaluate guardrail effectiveness and performance.
- Researched OWASP LLM security guidance and existing guardrail solutions to identify vulnerabilities and gaps in current LLM defenses.

REEL REVIEWS — Movie Review Web Application

December 2025

- Developed an interactive movie review web application with searchable movie content, ratings, reviews, and dynamic sorting functionality.
- Implemented JavaScript DOM manipulation, event handling, and localStorage to support user-submitted star ratings, reviews, comments, and persistent browser-side data.
- Processed movie data using Python and pandas, transforming IMDb dataset fields into JSON for integration with the front-end application.
- Built responsive movie pages using HTML and CSS and dynamically rendered movie information based on ratings and popularity.

Skills

Languages: Python, Java, JavaScript, HTML, CSS

Frameworks & Libraries: PyTorch, torchvision, NumPy, pandas, Matplotlib, SHAP

Security Tools: Rapid7, CrowdStrike, Cisco ISE, Zscaler (ZIA, ZPA, ZDX), Wireshark, OPNsense, Kali Linux

AI/LLM Security: LLM Guardrails, Prompt Injection & Jailbreak Detection, Adversarial Prompting, AI Safety, PII Detection, Data Leakage Prevention, Input/Output Filtering

Security & Risk: TPRM, Vendor Risk Assessment, Vulnerability Management, Risk Scoring, IOC-Based Detection, Signature-Based Detection, Zero Trust, DMZ & VPN Auditing

Security Frameworks: NIST CSF, NIST RMF, NIST SP 800-53, FFIEC

Developer Tools: GitHub, Google Colab, Jupyter Notebook, UTM, VirtualBox, Cursor, Claude Code, Copilot

Databases: Microsoft Access

Collaboration: Microsoft Teams, Zoom